GEOMETRIC POINT CRITERION FOR SET-THEORETIC EQUIVALENCE RELATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids-quotients.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The refinement (4) requires one of decent, locally separated, or quasi-separated; without these the equivalence of (4) can fail.

Editorial note. The label lemma-set-theoretic-equivalence-geometric names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Delivered card X2-alg-groupoids-quotients-set-theoretic-pre-equivalence-geometric (Lemma lemma-set-theoretic-pre-equivalence-geometric): the three-fold geometric criterion for j to be a set-theoretic pre-equivalence relation, including the containment of the images of the induced maps
- Chapter Definition definition-set-theoretic-equivalence, included as a context span below: a set-theoretic equivalence relation is a universally injective set-theoretic pre-equivalence relation
- Morphisms of Spaces, Lemma spaces-morphisms-lemma-universally-injective: universal injectivity can be tested by injectivity of the maps on k-valued points over algebraically closed fields
- More on Morphisms of Spaces, Lemma spaces-more-morphisms-lemma-when-universally-injective-radicial: a universally injective morphism that is decent, or locally separated, or quasi-separated is radicial
- Radicial morphisms: the fibres are spectra of purely inseparable field extensions, so over an algebraically closed field the reduction of a nonempty fibre is the spectrum of the base field itself

Source reference. Original equation 048S: sources/stacks-groupoids-quotients/pinned-original-source.tex, source lines 813–813. Prerequisite pointer invoked at 1230; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0491: sources/stacks-groupoids-quotients/pinned-original-source.tex, source lines 1091–1091. Prerequisite pointer invoked at 1246; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:groupoids-quotients.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0484: sources/stacks-chow/tags.txt, source lines 5531–5531. Prerequisite pointer invoked at 1259; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 040X: sources/stacks-chow/tags.txt, source lines 5272–5272. Prerequisite pointer invoked at 1250; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Definition 0.1. Let S be a scheme, and let B be an algebraic space over S. Let $j : R \to U \times_B U$ be a pre-relation over B.

(1) We say j is a *set-theoretic pre-equivalence relation* if for all algebraically closed fields k over B the relation $\sim_R$ on $U(k)$ defined by
$$\overline{u} \sim_R \overline{u}' \Leftrightarrow \begin{array}{c} \exists \text{ field extension } K/k,\ \exists\ r \in R(K), \\ s(r) = \overline{u},\ t(r) = \overline{u}' \end{array}$$
is an equivalence relation.

(2) We say j is a *set-theoretic equivalence relation* if j is universally injective and a set-theoretic pre-equivalence relation.

Lemma 0.2. *In the situation of Definition 0.1. The following are equivalent:*

(1) *The morphism j is a set-theoretic equivalence relation.*
(2) *The morphism j is universally injective and $j(|R|) \subset |U \times_B U|$ contains the image of $|j'|$ for any of the morphisms j' as in Equation (048S).*
(3) *For every algebraically closed field k over B of sufficiently large cardinality the map $j : R(k) \to U(k) \times U(k)$ is injective and its image is an equivalence relation.*

If j is decent, or locally separated, or quasi-separated these are also equivalent to

(4) *For every algebraically closed field k over B the map $j : R(k) \to U(k) \times U(k)$ is injective and its image is an equivalence relation.*

Proof. The implications (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (3) follow from Lemma 0491 and the definitions. The same lemma shows that (3) implies j is a set-theoretic pre-equivalence relation. But of course condition (3) also implies that j is universally injective, see Morphisms of Spaces, Lemma 040X, so that j is indeed a set-theoretic equivalence relation. At this point we know that (1), (2), (3) are all equivalent.

Condition (4) implies (3) without any further hypotheses on j. Assume j is decent, or locally separated, or quasi-separated and the equivalent conditions (1), (2), (3) hold. By More on Morphisms of Spaces, Lemma 0484 we see that j is radicial. Let k be any algebraically closed field over B. Let $\overline{u}, \overline{u}' \in U(k)$ with $\overline{u} \sim_R \overline{u}'$. We see that $R \times_{U \times_B U, (\overline{u}, \overline{u}')} \mathrm{Spec}(k)$ is nonempty. Hence, as j is radicial, its reduction is the spectrum of a field purely inseparable over k. As $k = \overline{k}$ we see that it is the spectrum of k. Whence a point $\overline{r} \in R(k)$ with $t(\overline{r}) = \overline{u}$ and $s(\overline{r}) = \overline{u}'$ as desired. $\square$